Universal formulas for CLF's with respect to Minkowski balls[1]

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Abstract

This note provides explicit, algebraic stabilizing formulas for clf's when controls are restricted to certain Minkowski balls in Euclidean space. Feedbacks of this kind are known to exist by a theorem of Artstein, but the proof of Artstein's theorem is nonconstructive. The formulas are used to construct approximation solutions to some stabilization problems.

1 Introduction

Smooth control-Lyapunov functions (clf's) provide foundations for much of current feedback control design. See for instance the appropriate sections in the textbooks [2, 6]. The theory of smooth clf's had its origins in Artstein's paper [1]. A very useful characteristic of clf's is the existence of 'universal formulas' for stabilization, as described in [5] and the above textbooks.

This note continues the search, started in [3] (and continued in [4]), for universal clf formulas for *constrained* controls. By a universal formula one means, informally (with precise definitions given later), an expression for a stabilizing feedback law in terms of the directional (or 'Lie') derivatives of the given clf (which is assumed known) in the directions of the vector fields that define the system. The paper [3] provided one such formula in the case of unit balls $\{x \in \mathbb{R}^m : \|x\|_2 < 1\}$ with respect to Euclidean norms. The formula is very simple. In fact, it is an algebraic function of the Lie derivatives. This paper will present analogous formulas for the case of control sets

$$B_{m,p} := \{x \in \mathbb{R}^m : \|x\|_p < 1\}$$

(unit balls in p-norms) for certain values of p other than two. These new formulas are then used to approximately solve the stabilization problem for the control sets $(-1,+1)^2$ and $B_{m,1}$.

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Consider the following control system on $\mathbb{R}^n$:

$$\dot{x} = f(x) + G(x)u \tag{1}$$

The entries of f and of the $n \times m$ matrix G are smooth functions on $\mathbb{R}^n$ and $f(0) = 0$. Controls take values in $B_{m,p}$, where p is a positive number that we further specify shortly. Suppose some feedback $k : \mathbb{R}^n \to B_{m,p}$ is such that

$$\dot{x} = f(x) + G(x)k(x) \tag{2}$$

has a continuous right side for $x \neq 0$ and is globally asymptotically stable about $x = 0$.

Converse Lyapunov theorems then guarantee the existence of a positive definite (meaning, $V(0) = 0$ and $V(x) > 0$ for $x \neq 0$), proper (i.e., $V(x) \to \infty$ as $\|x\|_2 \to \infty$), smooth mapping $V = V_k : \mathbb{R}^n \to \mathbb{R}$ such that

$$\inf_{u \in B_{m,p}} \{a(x) + b(x)\ u\} < 0 \tag{3}$$

for each $x \neq 0$, where we put $a(x) := \nabla V_k(x) f(x)$ and $b(x) := \nabla V_k(x) G(x)$. To show that V_k exists, simply find a Lyapunov function for (2) and put $u = k(x)$ in (3). Any positive definite, proper, smooth function satisfying (3) is called a *control-Lyapunov function (clf)* with respect to (1) with controls in $B_{m,p}$. If k is continuous at the origin, then V_k has an additional *small control property (scp):* For each $\varepsilon > 0$, there is a $\delta > 0$ such that if $x \neq 0$ satisfies $\|x\|_2 < \delta$, then there is some u with $\|u\|_2 < \varepsilon$ such that $a(x) + b(x)\ u < 0$. Observe, for future use, that a clf with respect to (1) and controls in $B_{m,p}$ having the scp is also a clf with respect to (1) and controls in W having the scp whenever $B_{m,p} \subseteq W$.

In [1], the following elegant converse to these facts is shown: If there is a clf V, then there is a feedback $k : \mathbb{R}^n \to B_{m,p}$ which is smooth on $\mathbb{R}^n \setminus \{0\}$ and which globally stabilizes the system, and this k can be taken to be continuous on $\mathbb{R}^n$ if V has the scp. The result holds for rather arbitrary control-value sets, including all Minkowski unit balls. The proof is nonconstructive, being based on partitions of unity.

In [3], an explicit formula for k which is algebraic in a and b is given for (1) for the case of controls valued in

$B_{m,2}$, but this formula is not appropriate for controls in other Minkowski balls.

In this paper, we give the following formulas for feedback stabilizers for controls valued in $B_{m,p}$ for p of the form $\frac{2r}{2r-1}$ and for any positive integers r and m:

$$k_p(x) = (k_{p,1}(x), \ldots, k_{p,m}(x)),$$

where, for $j = 1, \ldots, m$,

$$k_{p,j}(x) \;=\; \lambda_p\left(a(x), \|b(x)\|_{2r}^{2r}\right)(b_j(x))^{2r-1} \qquad (4)$$

and

$$\lambda_p(a,b) = -\begin{cases} \dfrac{a + \sqrt[2r]{a^{2r}+b^{2r}}}{\left(1+\sqrt[2r]{1+b^{2r-1}}\right)b} & \text{if } b > 0, \\ 0 & \text{if } b = 0. \end{cases}$$

Call a function on $\mathbb{R}^n$ *almost smooth* if it is smooth on $\mathbb{R}^n \setminus \{0\}$ and continuous. We generalize the construction in [3] by proving the following:

Theorem 1 *Let $p = \frac{2r}{2r-1}$ for some $r \in \mathbb{N}$. If V is a control-Lyapunov function with respect to (1) with controls in $B_{m,p}$, then the feedback given by (4) above is smooth on $\mathbb{R}^n \setminus \{0\}$, takes values in $B_{m,p}$, and globally stabilizes the system with respect to $x = 0$. Moreover, if the right-hand side of (1) is analytic in x and V is analytic, then k is analytic on $\mathbb{R}^n \setminus \{0\}$. Furthermore, if V satisfies the small control property, then the feedback is almost smooth on $\mathbb{R}^n$.*

Note that (4) with $r = 1$ gives the feedback stabilizer of [3]. Also note that the case of controls valued in the *closure* of the relevant Minkowski ball offers no difficulty, since if (3) holds with such controls, then by continuity it also holds with $u \in B_{m,p}$, so that (3) holding with the closed constraint set implies the existence of a feedback taking values in the ball itself (and hence in particular in its closure).

As an easy application of Theorem 1, we deduce the following approximation theorem. For each $S \subset \mathbb{R}^m$ and $\varepsilon > 0$, we put $S^\varepsilon := \{x \in \mathbb{R}^m : \inf_{y \in S} \|x - y\|_2 < \varepsilon\}$, the ε-enlargement of S.

Theorem 2 *Let $j = 1$ or ∞ (and, in the second case, assume $m = 2$). If V is a control-Lyapunov function satisfying the small control property for (1) with controls in $B_{m,j}$ and $\varepsilon > 0$ is given, then there is an almost smooth feedback ϕ_j, taking values in $B^\varepsilon_{m,j}$ and algebraic in the Lie derivatives, which globally stabilizes the system with respect to the equilibrium $x = 0$. This feedback is analytic on $\mathbb{R}^n \setminus \{0\}$ if the right side of (1) is analytic in x and V is analytic.*

In this way, algebraic feedback stabilization is also 'almost' possible for $B_{m,1}$ and $B_{2,\infty}$, modulo vanishingly small overflows of the feedbacks' values.

2 Main Lemmas

For any $U \subseteq \mathbb{R}^m$, let us denote by $\mathcal{D}(U)$ the set of $(a,b) \in \mathbb{R} \times \mathbb{R}^m$ such that $a + b \cdot u < 0$ for some $u \in U$. Recall the following definition:

Definition 2.1 Let $U \subseteq \mathbb{R}^m$. A **universal stabilizing formula** relative to U is a real-analytic function

$$\alpha : \mathcal{D}(U) \subseteq \mathbb{R} \times \mathbb{R}^m \to U \subseteq \mathbb{R}^m$$

satisfying these conditions:

1. For any (a,b) in $\mathcal{D}(U)$, $a + b \cdot \alpha(a,b) < 0$;
2. For each $\varepsilon > 0$, there exists a $\delta > 0$ such that if $(a,b) \in \mathcal{D}(U)$, $a < \delta\|b\|_2$, $|a| < \delta$, and $\|b\|_2 < \delta$, then $\|\alpha(a,b)\|_2 < \varepsilon$.

We require analyticity to disallow 'tricks' such as partitions of unity. As in [4], we reduce the problem of finding regular feedback laws to the problem of finding universal formulas. Notice that in condition 2. the norm $||\cdot||_2$ can be replaced by any equivalent norm. Moreover, we can replace condition 2. by the following:

2′. For each $\tilde{\varepsilon} > 0$ and any norms $\|\cdot\|_s, \|\cdot\|_t$, and $\|\cdot\|_u$ which are equivalent to $\|\cdot\|_2$, there is a $\tilde{\delta} > 0$ such that if (a,b) is in $\mathcal{D}(U)$, $a < \tilde{\delta}\|b\|_s$, $|a| < \tilde{\delta}$, and $\|b\|_t < \tilde{\delta}$, then $\|\alpha(a,b)\|_u < \tilde{\varepsilon}$.

From [4], we have the following:

Lemma 2.2 Let α be a universal stabilizing formula relative to $U \subseteq \mathbb{R}^m$. Then, for each system (1) and clf V relative to (1) and control set U, $k(x) := \alpha(a(x), b(x))$ is smooth on $\mathbb{R}^n \setminus \{0\}$ and globally stabilizes the system. (Here, $a(x) := \nabla V(x) f(x)$ and $b(x) := \nabla V(x) G(x)$.) If, in addition, V has the scp, then $\alpha(a(x), b(x))$ is almost smooth on $\mathbb{R}^n$. Moreover, if the right side of the system is analytic in x and V is analytic, then k is analytic on $\mathbb{R}^n \setminus \{0\}$.

The proof of the second theorem makes use of the following:

Lemma 2.3 If $(a,b) \to k(a,b)$ is a universal stabilizing formula relative to $U \subseteq \mathbb{R}^m$ and $T : \mathbb{R}^m \to \mathbb{R}^m$ is an invertible linear operator, then $k_T(a,b) := Tk(a, T'b)$ defines a universal stabilizing formula relative to TU.

Proof: Note that $\mathcal{D}(TU) = (I \times (T')^{-1})\mathcal{D}(U)$. Thus, if $(a,b) \in \mathcal{D}(TU)$, then $(a, T'b) \in \mathcal{D}(U)$, so

$$a + Tk(a, T'b) \cdot b = a + k(a, T'b) \cdot T'b < 0,$$

so k_T satisfies the first condition in Definition 2.1.

Now let $\varepsilon > 0$. Since $\|\cdot\|_2$ and $\|T'(\cdot)\|_2$ are equivalent, it remains to show that there is a $\delta > 0$ such that

$$\mathcal{D}(TU) \wedge \{a < \delta \|T'b\|_2,\ |a| < \delta,\ \|T'b\|_2 < \delta\}$$

$$\Rightarrow \|Tk(a, T'b)\|_2 < \varepsilon,$$

by the equivalence of conditions 2 and $2'$ above. But if $(a,b) \in \mathcal{D}(TU)$, then $(a, T'b) \in \mathcal{D}(U)$, so we can pick the δ such that

$$\mathcal{D}(U) \wedge \{a < \delta \|b\|_2,\ |a| < \delta,\ \|b\|_2 < \delta\}$$

$$\Rightarrow \|Tk(a,b)\|_2 < \varepsilon$$

to satisfy condition $2'$ (with $\|\cdot\|_s = \|\cdot\|_t = \|T'(\cdot)\|_2$ and $\|\cdot\|_u = \|\cdot\|_2$). ■

3 Proofs of the Theorems

To prove Theorem 1, it suffices to show that k_p, given by

$$k_{p,j}(a,b) \;=\; \lambda_p\left(a, \|b\|_{2r}^{2r}\right)\, b_j^{2r-1}$$

for $j = 1, \ldots, m$ is a universal stabilizing formula relative to $B_{m,p}$. The argument is analogous to the one presented for the $B_{m,2}$ case in [3]. We start with a simple characterization of $\mathcal{D}(B_{m,\frac{1}{s}})$.

Lemma 3.1 For all $s \in (0,1)$, let $\Gamma(s)$ denote the set $\{(a,b) \in \mathbb{R}\times\mathbb{R}^m : a < \|b\|_{\frac{1}{1-s}}\}$. Then, $\mathcal{D}(B_{m,1/s}) = \Gamma(s)$.

Proof: Fix $s \in (0,1)$. If $u \in B_{m,1/s}$ and if $a + b\cdot u < 0$, then Hölder's inequality gives

$$a \;<\; |b\cdot u| \;\le\; \|b\|_{\frac{1}{1-s}} \|u\|_{\frac{1}{s}} \;\le\; \|b\|_{\frac{1}{1-s}},$$

so $\mathcal{D}(B_{m,1/s}) \subseteq \Gamma(s)$. Conversely, if $a < \|b\|_{\frac{1}{1-s}}$, then we can find a $u \in B_{m,1/s}$ such that

$$b\cdot u \;=\; |b\cdot u| = \|b\|_{\frac{1}{1-s}} \|u\|_{\frac{1}{s}} \;=\; \|b\|_{\frac{1}{1-s}},$$

as follows. For $b = 0$, we choose $u = 0$; otherwise,

$$u_j := \left[\frac{|b_j|^{\frac{1}{1-s}}}{\|b\|_{\frac{1}{1-s}}^{\frac{1}{1-s}}}\right]^s \ \operatorname{sgn}(b_j) \quad \text{for } j = 1, \ldots, m.$$

Therefore $a < b\cdot u$ for some $u \in B_{m,1/s}$. This shows $\Gamma(s) \subseteq \mathcal{D}(B_{m,1/s})$ as well. ■

For each $n \in \mathbb{N}$, define the mapping $\tau = \tau_r$ by

$$(a,b) \longmapsto \begin{cases} \dfrac{a + \sqrt[2r]{a^{2r} + b^{2r}}}{b} & \text{if } b \neq 0 \\ 0 & \text{if } b = 0 \end{cases}$$

and put $S := \{(a,b) \in \mathbb{R}\times\mathbb{R} : b \le 0 \Rightarrow a < 0\}$. Then, for each $(a,b) \in S$, $(\tau(a,b), a, b)$ is a root of the polynomial

$$\begin{aligned} g(x,a,b) &= \frac{(bx-a)^{2r} - a^{2r} - b^{2r}}{b} \\ &= b^{2r-1}x^{2r} - 2rb^{2r-2}ax^{2r-1} + \cdots \\ &- 2ra^{2r-1}x - b^{2r-1}, \end{aligned}$$

and

$$\begin{aligned} \frac{\partial g}{\partial x}(\tau(a,b), a, b) &= 2r(b\tau(a,b) - a)^{2r-1} \\ &= 2r(a^{2r} + b^{2r})^{\frac{2r-1}{2r}} \end{aligned}$$

when $b \neq 0$. For $b = 0$, we get $-2ra^{2r-1}$. Since $\frac{\partial g}{\partial x}$ is never zero at these roots, we can conclude from the Implicit Function Theorem that τ is a real analytic function on the set S.

Now let p be as in the statement of our theorem. Observe that if $(a,b) \in \mathcal{D}(B_{m,p})$, with $a > 0$, then the inequality $a < \|b\|_{2r}$ established in Lemma 3.1 above (with $s = \frac{1}{p} = \frac{2r-1}{2r}$) gives

$$1 < \frac{1 + \sqrt[2r]{1 + (\|b\|_{2r}^{2r})^{2r-1}\,[\|b\|_{2r}^{2r}/a^{2r}]}}{1 + \sqrt[2r]{1 + (\|b\|_{2r}^{2r})^{2r-1}}},$$

i.e.,

$$a < \frac{a + \sqrt[2r]{a^{2r} + (\|b\|_{2r}^{2r})^{2r}}}{1 + \sqrt[2r]{1 + (\|b\|_{2r}^{2r})^{2r-1}}}\,. \tag{5}$$

If we instead had $a \le 0$, then (5) again holds, since $a + \sqrt[2r]{a^{2r} + (\|b\|_{2r}^{2r})^{2r}} \ge a + |a| = 0$. Therefore, k_p satisfies Condition 1 of Definition 2.1.

The composition of $-\tau$ and the analytic mapping

$$\mathcal{D}(B_{m,p}) \to S : (a,b) \longmapsto (a, \|b\|_{2r}^{2r})$$

is real analytic and $\sqrt[2r]{1 + b^{2r-1}}$ is real analytic and everywhere nonzero, so λ_p is real analytic, as is k_p.

Using once more the fact that $a^{2r} < \|b\|_{2r}^{2r}$ on $\mathcal{D}(B_{m,p})$ for $a > 0$ establishes that, on $\mathcal{D}(B_{m,p})$, the sum $\sum_{j=1}^m |k_{p,j}|^{\frac{2r}{2r-1}}$ is majorized by

$$\left\{\frac{||b||_{2r}\left[1 + \sqrt[2r]{1 + [\|b\|_{2r}^{2r})^{2r-1}}\right]}{\|b\|_{2r}^{2r}\left[1 + \sqrt[2r]{1 + [\|b\|_{2r}^{2r}]^{2r-1}}\right]}\right\}^{\frac{2r}{2r-1}} \sum_{j=1}^m b_j^{2r},$$

which is at most $\frac{\|b\|_{2r}^{2r}}{\|b\|_{2r}^{2r}} = 1$. For $a \le 0$, the majorizer is

$$\left\{\frac{\|b\|_{2r}^{2r}}{\left[1 + \sqrt[2r]{1 + [\|b\|_{2r}^{2r}]^{2r-1}}\right]\|b\|_{2r}^{2r}}\right\}^{\frac{2r}{2r-1}} \sum_{j=1}^m b_j^{2r}$$

which, for $b \neq 0$, is

$$< \left[\frac{\|b\|_{2r}^{2r}}{[\|b\|_{2r}^{2r}]^{1-\frac{1}{2r}} \|b\|_{2r}^{2r}} \right]^{\frac{2r}{2r-1}} \|b\|_{2r}^{2r} = 1.$$

In the first case, the majorization is a strict one, so we conclude that k_p is valued in $B_{m,p}$.

Conditions 2 and $2'$ being equivalent, our first theorem therefore follows if, given $\varepsilon > 0$, one could find a $\delta > 0$ such that $\|k_p\|_p < \varepsilon$ whenever

$$\mathcal{D}(B_{m,p}) \wedge (a < \delta \|b\|_{2r},\ |a| < \delta,\ \|b\|_{2r} < \delta)$$

We show this next. (The fact that $(a,b) \in \mathcal{D}(B_{m,p})$ is redundant.) Let (a,b) be as in the hypothesis of this implication, with $\delta \in (0,1)$ arbitrary for the moment. Then, for $b \neq 0$, $\|k_p\|_p$ is majorized by

$$\frac{\delta + \sqrt[2r]{\delta^{2r} + (\|b\|_{2r}^{2r})^{2r-1}}}{1 + \sqrt[2r]{1 + (\|b\|_{2r}^{2r})^{2r-1}}} ,$$

which can evidently be made as small as desired when $\|b\|_{2r}$ and δ are both taken to be small enough, so the result holds.

To prove our second theorem, let $\varepsilon > 0$ be given, then pick $p = \frac{2r}{2r-1}$ with r large enough so that $B_{m,p} \subset B_{m,1}^{\varepsilon}$. As k_p is a universal stabilizing formula for $B_{m,p}$, and since V is a clf for $B_{m,p}$ when it is one for $B_{m,1}$, we can pick $\phi_1 = k_p$.

The result for the $B_{m,\infty}$ case with $m = 2$ follows by rotating the $p = 1$ result. Let T be the rotation-dilation of $\mathbb{R}^m$ satisfying $TB_{2,1} = B_{2,\infty}$, and let $\varepsilon > 0$ be given. Let ϕ_1 be the version associated with $\varepsilon/\sqrt{2}$. That we can set $\phi_\infty = T\phi_1(a, T'b)$ is easily checked using Lemma 2.3.

4 Remarks

In this section, we show how to view the feedbacks in (4) as members of a larger family of explicit, smooth, not-necessarily algebraic explicit feedback laws. Let us introduce the following

Definition 4.1 A **universal stability map (usm)** is a log concave function $\theta : [0,\infty) \to [0,\infty)$ which is real analytic on $(0,\infty)$ and satisfies $\theta(xy) \geq \theta(x)\theta(y)$ for all x and y in $[0,\infty)$ and $\theta'(x) > 0$ for all $x \in (0,\infty)$. A usm is called **invertible** if its inverse extends to an even real analytic function on $\mathbb{R}$.

The mapping $x \longmapsto x^{\frac{1}{2r}}$ is an invertible usm for any $r \in N$, but there are nonalgebraic invertible usm's also. Here is an example:

Remark 4.2 The function $x \stackrel{\alpha}{\longmapsto} (x \cdot \tanh(x))^{\frac{1}{4}}$ is an invertible usm.

The log concavity of α and the fact that $\alpha' > 0$ on $(0,\infty)$ are easily checked. Since $\tanh(\cdot)$ is real analytic, we conclude that α is a usm as long as $\tanh(xy) \geq \tanh(x)\tanh(y)$ for x and y in $(0,\infty)$. If $y \geq 1$, the monotonicity of tanh implies that $\tanh(y) \leq 1 \leq \tanh(xy)/\tanh(x)$, so by symmetry in x and y we can assume that x and y are both smaller than one. Fix $y \in (0,1)$, and define θ_y by $\theta_y(x) := \tanh(xy) - \tanh(x) \cdot \tanh(y)$. It remains to show that $\theta_y'(x) \geq 0$ for all $x \in (0,1)$, i.e., that

$$\left(\frac{\cosh(x)}{\cosh(xy)} \right)^2 \geq \frac{\tanh(y)}{y} \tag{6}$$

But if $y < 1$, then $x > xy$, so the left of (6) is at least one, and one dominates the right side, since $\tanh(y) - y$ has nonpositive derivative $1/\cosh^2(y) - 1$. The invertibility of α will follow from the following:

Remark 4.3 Assume $f(x) = \sqrt{x}\ [1 + h(x)]$ for some real analytic $h : (-\varepsilon, \varepsilon) \to \mathbb{R}$ with $h(0) = 0$ (for some $\varepsilon > 0$). Then there is an even function K, defined and real analytic near 0, whose restriction to a neighborhood of zero inverts f.

Proof: Introduce $g(u) = u[1 + h(u^2)]$. Since $g'(0) \neq 0$, we can let G denote the local inverse of g near 0, the existence of which is guaranteed by the Inverse Function Theorem. Then, $G(g(\sqrt{x}\,)) = \sqrt{x}$ for $x > 0$ near zero, and $G[\sqrt{x}\,(1 + h(x))] = \sqrt{x}$, so $x = [G(f(x))]^2$. Put $K(y) = [G(y)]^2$, which is analytic near 0. Since g is odd, G is odd, so K is even. ■

Returning to Remark 4.2, note that we can write $x \tanh(x) = x^2(1 + \kappa(x))$ for a suitable analytic function κ, which gives $\alpha(x) = \sqrt{x}\,(1 + h(x))$ if we put $h(x) = \sqrt[4]{1 + \kappa(x)} - 1$. Note that $\kappa(0) = 0$, so $h(0) = 0$; and h is analytic near 0. Applying Remark 4.3, we conclude that $\alpha(x)$ has an analytic inverse K extending to an even function in some neighborhood of 0. But, $\alpha(x)$ has an analytic inverse about any positive number, by the Inverse Function Theorem.

By the uniqueness of inverses, α^{-1} therefore extends through the positive numbers to a real analytic $\tilde{\gamma}$. Extend this $\tilde{\gamma}$ by symmetry with respect to $x = 0$ to get a function γ defined on all of $\mathbb{R}$. This γ will be real analytic, since on $(-\infty, 0)$ it is the composition of $x \longmapsto -x$ and the restriction of $\tilde{\gamma}$ to the positive reals and it equals K near zero. Therefore, γ inverts α. ■

From [5], we know that the function defined on the set $S := \{(a,b) \in \mathbb{R}^2 : b \leq 0 \Rightarrow a < 0\}$ by

$$(a,b) \longmapsto \begin{cases} \dfrac{a + \sqrt{a^2 + b^4}}{b} & \text{if } b \neq 0 \\ 0 & \text{if } b = 0 \end{cases}$$

is real analytic. Here is a generalization:

Remark 4.4 Let S be as above, let θ be an invertible usm, and let θ^{-1} extend to the even analytic function γ on $\mathbb{R}$. Let $\zeta : \mathbb{R} \to \mathbb{R}$ be a positive definite, real analytic function and define $\phi : S \to \mathbb{R}$ by

$$\phi(a,b) = \begin{cases} \dfrac{\theta\left(\gamma(a)+\zeta(b)b^2\right)+a}{b} & \text{if } b \neq 0 \\ 0 & \text{if } b = 0 \end{cases}$$

Then ϕ is real analytic.

Proof: Note that if $f : \mathbb{R}^3 \to \mathbb{R}$ is real analytic, then

$$D^f(a,t,y,b) = \begin{cases} \dfrac{f(a,t+b,y)-f(a,t,y)}{b}, & b \neq 0 \\ \frac{\partial f}{\partial t}(a,t,y), & b = 0 \end{cases}$$

is a real analytic function on $\mathbb{R}^4$, as is seen by writing it as $\int_0^1 \frac{\partial f}{\partial t}(a, t+\lambda b, y)\, d\lambda$. Putting $f(a,t,y) := \gamma(ty-a)$ and then evaluating the result at $t=0$ shows that

$$L(y,a,b) := \begin{cases} \dfrac{\gamma(by-a)-\gamma(a)-\zeta(b)b^2}{b}, & b \neq 0 \\ y\gamma'(-a), & b = 0 \end{cases}$$

too is real analytic. But, $(\phi(a,b),a,b)$ is a root of L for all $(a,b) \in S$, so analyticity will follow from the Implicit Function Theorem if $\frac{\partial L}{\partial y}(\phi(a,b),a,b) \neq 0$ on S. That this is indeed this case when $b \neq 0$ follows from the fact that γ' and θ are positive on $(0,\infty)$. For $b=0$, we use the facts that a and b are not simultaneously zero on S and $|\gamma'(\cdot)|$ is positive away from zero. ∎

The following is a consequence:

Remark 4.5 Let $\lambda : \mathbb{R}^m \to \mathbb{R}_{\geq 0}$ be scalar homogeneous (i.e., assume $\lambda(\alpha x) = \alpha\lambda(x)$ for all $\alpha \geq 0$ and $x \in \mathbb{R}^m$), and put $U := \{\lambda(\cdot) < 1\}$. Assume there are an invertible usm θ through the origin and a real analytic function $q : \mathbb{R} \to \mathbb{R}$ such that

$$\left\{(a,b) \in \mathbb{R}\times\mathbb{R}^m : a < \theta\left(\ \sum_{j=1}^m q(b_j)b_j\right)\right\}$$

contains $\mathcal{D}(U)$. Let ζ and γ be as in Remark 4.4, with β denoting the mapping $b \longmapsto \sum_{j=1}^m q(b_j)b_j$. Define Q by $Q(b) = (q(b_1,\ldots,b_m))$. If

$$\lambda(Q(b)) \leq \frac{[1+\theta\{1+\zeta[\beta(b)]\beta(b)\}]\,\beta(b)}{\theta\{\beta(b)\}+\theta\{\beta(b)+\zeta[\beta(b)][\beta(b)]^2\}}$$

for all $(a,b) \in \mathcal{D}(U)$ with $b \neq 0$, then the mapping $\mu = (\mu_1,\ldots,\mu_m)$ defined by

$$\mu_j(a,b) := -\frac{a+\theta\left(\gamma(a)+\zeta[\beta(b)][\beta(b)]^2\right)}{(1+\theta\{1+\zeta[\beta(b)]\beta(b)\})\beta(b)}q(b_j)$$

for $b \neq 0$ and 0 if $b=0$ is a real analytic function on $\mathcal{D}(U)$ satisfying $a < \mu(a,b)\cdot b$ for all $(a,b) \in \mathcal{D}(U)$. It is valued in U if θ is subadditive.

In the last case, if V is a clf with respect to (1) and U, then, with $a := \nabla V(x)f(x)$ and $b := \nabla V(x)G(x)$, μ is smooth on $\mathbb{R}^n \setminus \{0\}$ and globally stabilizes the system with respect to $x=0$. If the right side of (1) is analytic in x and V is analytic, then μ is analytic on $\mathbb{R}^N \setminus \{0\}$.

The proof is similar to the proof we gave in the previous section. With $\lambda(x) = \|x\|_{2r/(2r-1)}$, $\theta(x) = x^{1/2r}$, and $q(x) = x^{2r-1}$, we get back the result of Theorem 1.

5 Conclusions

Let $\varepsilon > 0$ be given, and let V be an analytic clf for (1) with controls in $(-1,+1)^2$ which has the scp. Assume f and G are real analytic. One can find a feedback $k : D((-1,+1)^2) \to ((-1,+1)^2)^\varepsilon$ such that (2) has $x=0$ as a globally stable equilibrium. The feedback is a simple algebraic function of the Lie derivatives and is analytic on $\mathbb{R}^n \setminus \{0\}$. The same is true with $(-1,+1)^2$ replaced by $B_{m,1}$ for any m in $\mathbb{N}$. If V is smooth, then k is almost smooth.

In both cases, the formulas are members of a family of algebraic universal stabilizing formulas having the feedback formula of [3] as one of its members, up to a rotation. The formulas are members of a general class of explicit nonalgebraic feedback laws with similar properties. Thus, we can stabilize (1) relative to $U = B_{m,1}$ or $U = B_{2,\infty}$ with an algebraic feedback k as long as we permit k to take its values in U^ε, where $\varepsilon > 0$ is as small as desired. We can also globally stabilize (1) with respect to $U := B_{m,p}$ when $p = \frac{2r}{2r-1}$ and $m, r \in \mathbb{N}$ using universal stabilizing formulas. This generalizes [3], which treats only the case $U = B_{m,2}$.

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